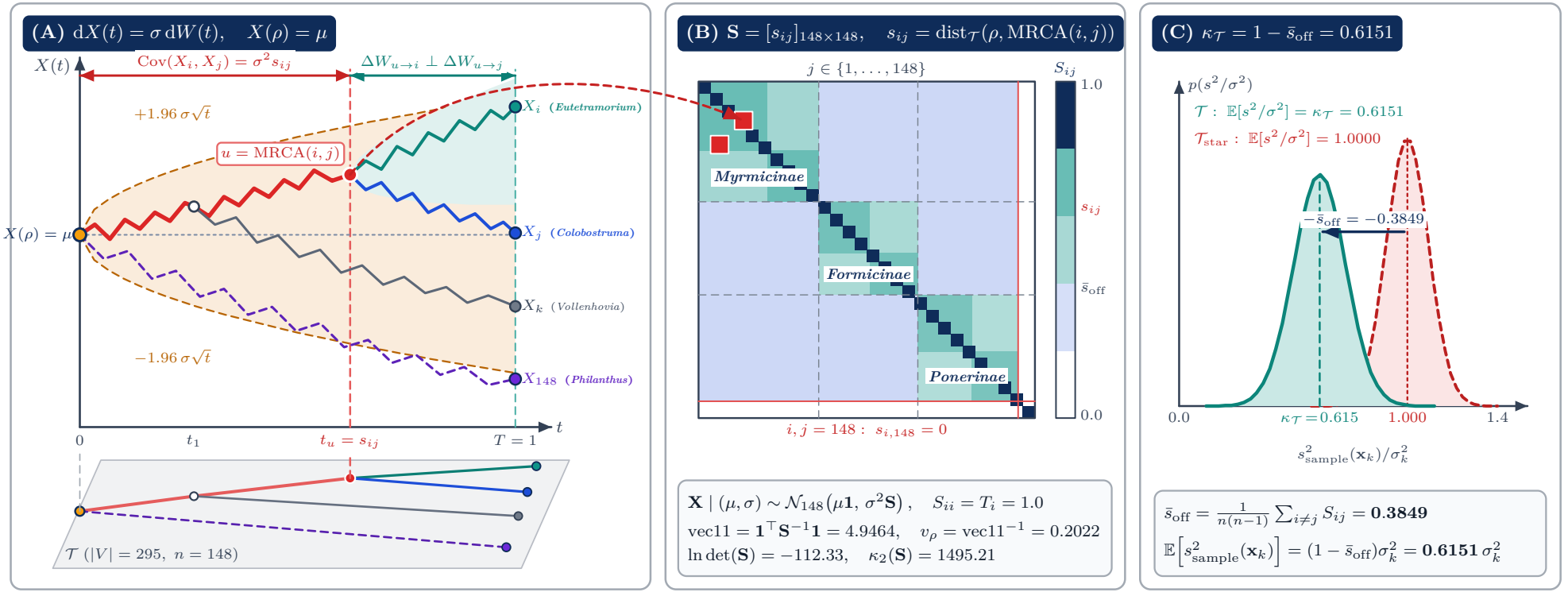

(A) $\mathrm{d}X(t) = \sigma\,\mathrm{d}W(t), \quad X(\rho) = \mu$
$X(t)$
$\mathrm{Cov}(X_i, X_j) = \sigma^2 s_{ij}$
$\Delta W_{u\to i} \perp \Delta W_{u\to j}$
$+1.96\,\sigma\sqrt{t}$
$u = \mathrm{MRCA}(i,j)$
X_i (Eutetramorium)
$X(\rho) = \mu$
X_j (Colobostruma)
X_k (Vollenhovia)
$-1.96\,\sigma\sqrt{t}$
X_{148} (Philanthus)
t
0
t_1
$t_u = s_{ij}$
$T = 1$
$\mathcal{T}$ ($|V| = 295$, $n = 148$)
(B) $\mathbf{S} = [s_{ij}]_{148\times148}, \quad s_{ij} = \mathrm{dist}_{\mathcal{T}}(\rho, \mathrm{MRCA}(i,j))$
$j \in \{1, \ldots, 148\}$
S_{ij}
1.0
s_{ij}
$\bar{s}_{\mathrm{off}}$
0.0
Myrmicinae
Formicinae
Ponerinae
$i, j = 148: \; s_{i,148} = 0$
$\mathbf{X} \mid (\mu, \sigma) \sim \mathcal{N}_{148}(\mu\mathbf{1}, \sigma^2\mathbf{S}), \quad S_{ii} = T_i = 1.0$
$\mathrm{vec11} = \mathbf{1}^\top\mathbf{S}^{-1}\mathbf{1} = 4.9464, \quad v_\rho = \mathrm{vec11}^{-1} = 0.2022$
$\ln\det(\mathbf{S}) = -112.33, \quad \kappa_2(\mathbf{S}) = 1495.21$
(C) $\kappa_{\mathcal{T}} = 1 - \bar{s}_{\mathrm{off}} = 0.6151$
$p(s^2/\sigma^2)$
$\mathcal{T}$: $\mathbb{E}[s^2/\sigma^2] = \kappa_{\mathcal{T}} = 0.6151$
$\mathcal{T}_{\mathrm{star}}$: $\mathbb{E}[s^2/\sigma^2] = 1.0000$
$-\bar{s}_{\mathrm{off}} = -0.3849$
0.0
$\kappa_{\mathcal{T}} = 0.615$
1.000
1.4
$s^2_{\mathrm{sample}}(\mathbf{x}_k)/\sigma^2_k$
$\bar{s}_{\mathrm{off}} = \frac{1}{n(n-1)}\sum_{i\neq j} S_{ij} = \mathbf{0.3849}$
$\mathbb{E}\left[s^2_{\mathrm{sample}}(\mathbf{x}_k)\right] = (1 - \bar{s}_{\mathrm{off}})\sigma^2_k = \mathbf{0.6151}\,\sigma^2_k$